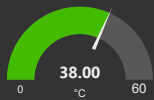


Digital Twin Intelligence

Process Overview

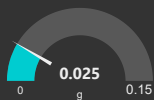
Temperature (°C)



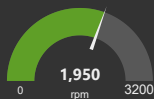
Motor Current (A)



Vibration (g RMS)



Fan Speed (RPM)



Power (W)



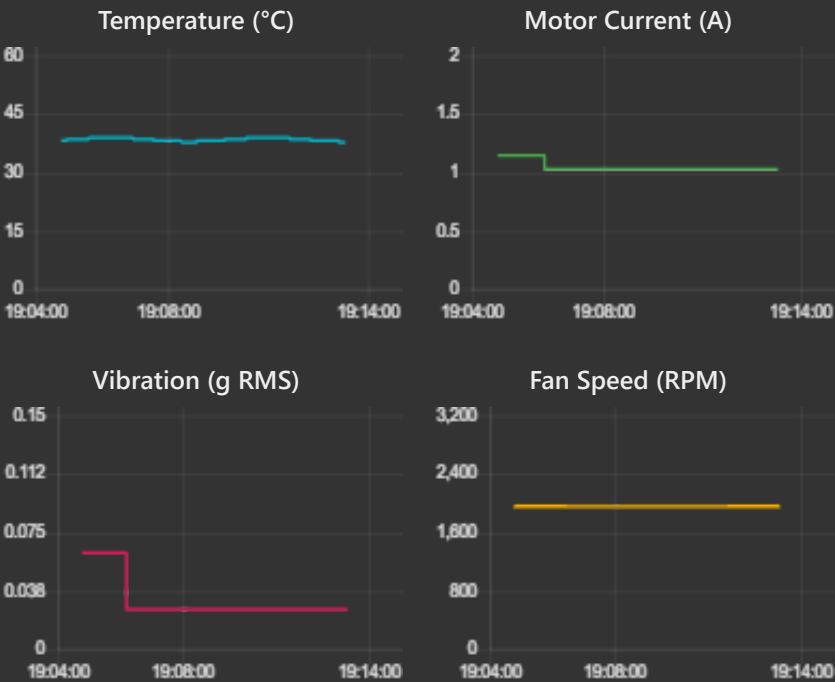
Digital Twin State

SustainTwin Research Demonstrator

Physics-aware digital twin • Controlled fault scenarios • 10-minute experiments • Sustainability analytics • AI inference

Asset	FAN_DT_001	Model	process-model-3.0.0-build3	Schema
Control / Operation	AUTOMATIC NORMAL		Health / Fault	HEALTHY F0

Performance Trends



Supervisory Control

Control: Auto

Pending Command: AUTO W

HIGH LOAD

Sustainability Insights



Fault Experiment

Fault Scenario: F1 — Mechanical imbalance Fault Severity: 0.6 Replicate: 1

Selected Scenario: F1 | Severity 0.6 | Replicate 1

START 10-MIN EXPERIMENT STOP / ABORT RUN Phase: COMPLETE

Experiment Status: COMPLETE — 10-minute CSV saved | FAN_DT_001_F1_S06_R01_20260826T175712

Ground Truth	F0	Fault Label	healthy	Fault Severity

Closed-Loop Status

Control Mode	AUTOMATIC	Operating Mode	NORMAL	Speed Set-point	65 %	Target Speed	1,950 rpm	Actual Speed	1,950 rpm
Command Status	ACCEPTED	Approx. Response	673 ms	Commands	8 accepted / 0 rejected				

Last Command Sent	<code>{"clear_fault":true}</code>
Asset Health	HEALTHY F0